Jio Platforms Limited

WSO2 API MANAGER SETUP GUIDE VERSION 4.0.0

Consent Management System

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Version History

Date	Contributor	Contribution	Version	State
Nov 25, 2025	Racharla Vamshi	Initial Draft	0.1	Draft
Nov 26, 2025	Lokesh Sharma	Reviewed and released the document	1.0	Released

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1. Overview

This guide provides **step-by-step instructions** for deploying the Jio Consent Management System (JCMS). It is designed for both technical and non-technical users.

Technology Stack: - **Backend:** Spring Boot (Java 21) - **Frontend:** ReactJS, Node.js 20 - **Other:** Docker, Kubernetes (with ConfigMaps), MongoDB v7.0.14, Nginx - **Open-Source Components:** WSO2 v4.0.0, HashiCorp Vault v1.19.4, Nginx with Lua module, Apache Kafka (open source)

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Purpose: Secure, compliant, and scalable consent and grievance management for Data Fiduciaries and Data Principals.

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2. Open-Source Setup & Credential Management

2.1. Open-Source Component Deployment

- **WSO2 Setup:**
  - Navigate to `JCMS/OpenSourceConfiguration/Wso2/`
  - Follow the README guide
  - Note down service URLs and ports
- **HashiCorp Vault Setup:**
  - Navigate to `JCMS/OpenSourceConfiguration/HashiCorpVault/`
  - Deploy Vault and configure secrets storage
  - Store private certificates, AES keys, and service credentials
  - Verify Vault URLs and token accessibility
- **Nginx Deployment:**
  - Deploy Nginx server with Lua module
  - Configure Lua scripts for consent and grievance endpoints

2.2. Credential Preparation

Credential Type	Purpose	Location / Notes
MongoDB Standard User	Backend DB access	`mongodb-secret`
MongoDB Admin User	Admin operations	`admin-mongodb-secret`
WSO2 Credentials	API access	Generated via WSO2 Cred Generator

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Credential Type	Purpose	Location / Notes
Vault Token / Keys	Certificates & encryption	Accessible via Vault service
JWT / Auth Secrets	User & business authentication	Configured per microservice
Bhashini Translation Credentials	Multi-language support	Configured per translator API
DigiLocker Credentials	DigiLocker API integration	Stored securely in backend services

1. Environment Variables

```
NAMESPACE=<your_k8s_namespace>
```

WSO2 API Manager is an open-source full-lifecycle API management platform.

It supports:

- API onboarding & publishing
- API lifecycle management (create, test, publish, deprecate)
- Subscriptions & access control
- Throttling & rate limiting
- API security (O
- Auth2/JWT, mTLS, key validation)
- Traffic management & gateway routing
- Multi-tenant support
- Analytics for traffic, faults, latency, users, tenants & businesses
- Developer Portal for client onboarding
- Publisher Portal for API creation and deployment

2. Project Usage

In this project, WSO2 API Manager is used mainly for:

- Capturing analytics of API usage segregated by Tenant and Business.

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- Securely exposing RP-APIs to businesses (automated through Partner Portal).

3. Prerequisites

1.A suitable VM (Linux recommended)
  - Minimum 2 GB RAM
  - 4 CPU cores
  - At least 10 GB disk

2.Java 8 (required for WSO2 4.0.0)

Documentation: https://apim.docs.wso2.com/en/4.0.0/install-and-setup/install/installation-prerequisites/

4. Setup Instructions

- Download WSO2 API Manager 4.0.0
  - URL: https://github.com/wso2/product-apim/releases/download/v4.0.0/wso2am-4.0.0.zip
- Extract the ZIP
  - Example path: /app/wso2/
  - After extraction you will see: wso2am-4.0.0/
- Navigate to the bin directory
  - cd /app/wso2/wso2am-4.0.0/bin
- Start WSO2 API Manager
  - Command:
    - ./api-manager.sh
    - To run in background:
    - nohup ./api-manager.sh 2>&1 &
- Verify the server startup
  - Access the URL:
    - https://<ip-or-host>:9443/publisher
    - Default credentials:
    - admin / admin
- Optional Recommended Configurations
  - Configure MySQL/PostgreSQL instead of default H2

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- Update keystore & SSL certificates
- Change all default passwords
- Enable password encryption for sensitive configs
- Configure analytics & dashboards
- Enable distributed deployment if required (Gateway + Key Manager + Analytics)

Documentation: https://apim.docs.wso2.com/en/4.0.0/

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5. Notes for Easier Client Usage

• Always run under a non-root user
• Keep JAVA_HOME correctly set before startup
• Expose ports 9443 (console), 8280/8243 (API Gateway)
• Use nohup or systemd to keep the service running after logout
• Copy the log path for debugging:
/app/wso2/wso2am-4.0.0/repository/logs/wso2carbon.log

6. Integration With WSO2 Wrapper APIs

- This wso2 is supposed to be onboarding all the APIs of NeGD.(follow the official documentation to onboard it)
- For each API onboarded, the apiId should be put in the db manually for WSO2 Wrapper APIs for Business to subscribe automatically.

7. How to get the API Id:

- Log in to the **Publisher Portal**.

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- Go to the **APIs** list.

- Click on the API you want.

- Look at the **browser URL** at the top.
- It will look like: https://<host>/publisher/apis/<API_ID>/overview
- The part between **/apis/** and **/overview** is the **API ID**.
- Example: https://host/publisher/apis/bda997a3-dce4-43dd-a2bd-ee3b4c3fdd7a/overview
  Here: API ID = bda997a3-dce4-43dd-a2bd-ee3b4c3fdd7a
- Same method works on **DevPortal**:

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